**University name**
Faculty / Department

**Operating Systems**
Exercise 3 – Hyper-V virtualiser and comparison of virtualisers available on the market

# Type 1 and Type 2 Hypervisors

Definitions, differences, advantages and disadvantages, applications, and an overview of Type 1, Type 2 and server-designed virtualisers

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# 1. Introduction

**Virtualisation** is a technique that separates software from the physical hardware it runs on. A thin layer of software creates an abstraction of the processor, memory, storage and network devices, so that several isolated **virtual machines (VMs)** – each with its own operating system (the **guest OS**) – can run at the same time on one physical computer (the **host**).

The component that makes this possible is the **hypervisor**, also called the **virtual machine monitor (VMM)**. Its tasks are: allocating CPU time, memory and I/O devices to the virtual machines, isolating the VMs from each other and from the host, intercepting privileged instructions issued by guest operating systems, and controlling communication between the VMs and the physical hardware.

The classification of hypervisors into **Type 1** and **Type 2** was introduced by Robert P. Goldberg in 1973 [3] and is still the standard way of describing them. The criterion is simple: **where the hypervisor sits relative to the hardware and the operating system**. A Type 1 hypervisor runs directly on the hardware, while a Type 2 hypervisor runs as an application on top of an existing operating system (Figure 1).

Modern x86 processors include hardware support for virtualisation – **Intel VT-x** and **AMD-V** (CPU virtualisation), **EPT/NPT** (also called SLAT – hardware translation of guest memory addresses) and **VT-d/AMD-Vi** (IOMMU, used to assign physical devices directly to a VM). These extensions are required by most current Type 1 hypervisors and are also used by Type 2 hypervisors, which greatly reduces the performance gap that existed when virtualisation relied only on software techniques such as binary translation [1][2].

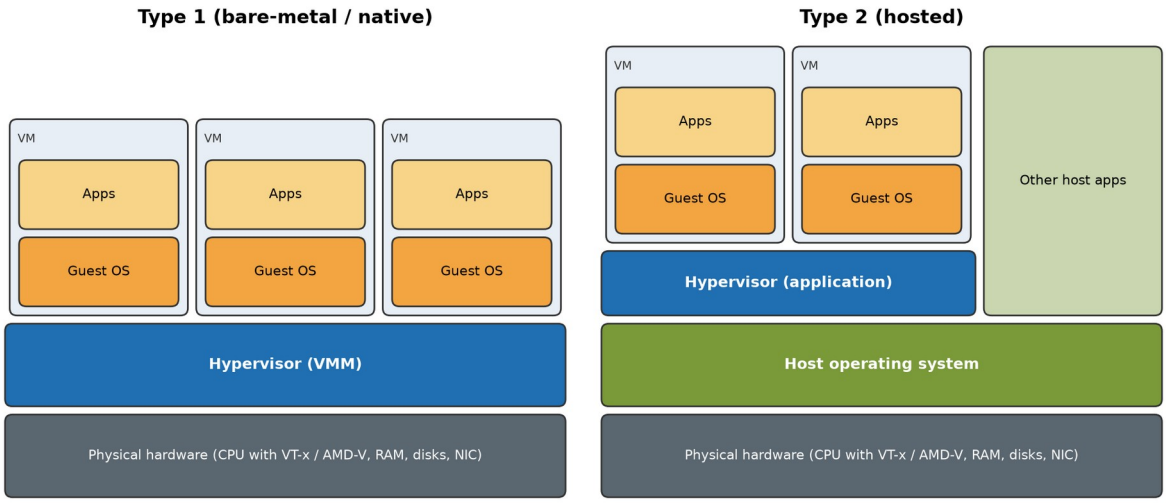



*Figure 1. Architecture of a Type 1 (left) and a Type 2 (right) hypervisor. Own work based on [1].*

# 2. Type 1 hypervisor (bare-metal, native)

## 2.1. Definition

A **Type 1 hypervisor** is installed **directly on the physical hardware** ("bare metal") instead of on an operating system. It is itself a specialised, minimal operating system whose only job is to run virtual machines. It has **direct access to hardware resources**, without an intermediate general-purpose operating system and its drivers. It uses full virtualisation with hardware acceleration, which it needs to work correctly [1].

In some designs a privileged management VM sits beside the hypervisor and supplies the device drivers and management tools: **dom0** in Xen and the **root (parent) partition** in Microsoft Hyper-V. Even then the hypervisor runs underneath all operating systems, including the management one, so the design is still Type 1.

## 2.2. Capabilities

- Near-native performance, because no host OS sits between the VMs and the hardware.
- Running dozens or hundreds of VMs on one server (server consolidation).
- Advanced memory management: overcommitment, page sharing, ballooning and compression.
- Live migration of a running VM between physical hosts without stopping it (e.g. vMotion, Hyper-V Live Migration).
- High availability (HA) and clustering: VMs restart automatically on another host after a failure.
- Direct hardware assignment: PCI passthrough, SR-IOV network cards, GPU sharing or passthrough (e.g. for VDI or AI workloads).
- Remote, centralised management of many hosts through web consoles and APIs.
- Strong isolation and a small attack surface, because the hypervisor's code base is small.

## 2.3. Limitations

- **Limited hardware support.** No general-purpose OS supplies drivers, so the hypervisor runs only on certified hardware listed in its hardware compatibility list (HCL), which is especially true for VMware ESXi [1].
- **Needs a dedicated machine.** The hypervisor replaces the operating system, so the machine cannot also serve as an ordinary desktop.

- **Harder to install and administer.** It requires knowledge of networking, storage and remote management tools.
- **Licensing cost.** Commercial enterprise editions (VMware vSphere, Windows Server Datacenter) can be expensive, although free open-source options exist (KVM, Xen, Proxmox VE, XCP-ng).
- **Usually managed from another computer.** Day-to-day management needs a separate computer with a browser or console.

## 2.4. Applications

- Data centres and server rooms: consolidating many physical servers onto a few hosts.
- Public and private clouds (IaaS): AWS (the Nitro hypervisor is based on KVM), Google Cloud (KVM), Microsoft Azure (Hyper-V), and OpenStack-based clouds.
- Production company workloads: domain controllers, databases, mail, ERP and file servers.
- Virtual Desktop Infrastructure (VDI): many user desktops served from central servers.
- Hosting and VPS providers, telecoms (NFV – network function virtualisation) and edge computing.
- Disaster recovery: replicating VMs to a second site (e.g. Hyper-V Replica, vSphere Replication).

*Table 1. Advantages and disadvantages of Type 1 hypervisors.*

| Advantages | Disadvantages |
|---|---|
| • Highest performance and efficient use of resources<br>• Better security: small attack surface, does not inherit the vulnerabilities of a desktop OS<br>• Stability and reliability – designed for 24/7 operation<br>• Scalability: many VMs per host, clusters of many hosts<br>• Enterprise features: live migration, HA, load balancing, replication<br>• Centralised remote management and automation (APIs, PowerShell, Ansible) | • Limited range of supported hardware (HCL)<br>• Needs a dedicated physical machine<br>• More complex installation, configuration and administration<br>• Commercial licences can be expensive<br>• Requires a separate management workstation or console<br>• A misconfiguration or failure of the host affects all of its VMs |

# 3. Type 2 hypervisor (hosted)

## 3.1. Definition

A **Type 2 hypervisor** is an **application installed on an existing operating system** (the host OS), just like any other program. It is **fully dependent on the host OS**, which it uses to

manage CPU scheduling, RAM, disk reads and writes, and access to devices. Because the host OS handles the hardware, the hypervisor can run on almost any computer that the OS supports. The cost is lower performance, since every guest operation passes through an additional layer [1].

Type 2 hypervisors also use hardware acceleration (VT-x/AMD-V) when it is available. Some of them can fall back to software techniques such as binary translation or full emulation (QEMU TCG) when the hardware lacks the needed features [1].

## 3.2. Capabilities

- Quick installation – a few minutes, like any other application; no dedicated machine needed.
- Runs on ordinary laptops and desktops with Windows, macOS or Linux, so it supports a wide range of hardware.
- Host and guest systems work at the same time, side by side, on one screen.
- Desktop integration features: shared folders, a shared clipboard, drag-and-drop, USB device redirection, seamless or unity mode (guest application windows on the host desktop) and 3D graphics acceleration.
- Snapshots and clones, so it is easy to go back to a known state – ideal for experiments.
- Flexible virtual networking: NAT, bridged, host-only and internal networks.
- Some products (QEMU) can emulate a different CPU architecture, e.g. ARM or RISC-V on an x86 PC.

## 3.3. Limitations

- **Lower performance**, especially for disk and network I/O, because requests pass through the host OS and its drivers. VMs also compete with host applications for CPU and RAM.
- **Security depends on the host OS.** If the host is compromised, all VMs are compromised. The attack surface is larger (host OS plus hypervisor application). Vulnerabilities in emulated devices, such as VENOM (CVE-2015-3456) in QEMU's virtual floppy controller, can allow a "VM escape".
- **Stability depends on the host.** Host OS updates, restarts or crashes stop every VM.
- **Limited scalability.** Typically a few VMs per host, with no clustering, HA or live migration between hosts.

- **Conflicts between hypervisors.** On Windows, enabling Hyper-V (also used by WSL 2 and virtualisation-based security) means that VirtualBox and VMware Workstation must run through the Windows Hypervisor Platform API, which can reduce their performance.
- **Not suitable for production servers** that must run 24/7.

### 3.4. Applications

- Education: learning operating systems (e.g. installing Linux on a Windows laptop for OS classes) and building lab networks.
- Software development and testing on several operating systems or versions, and reproducing bugs.
- Running applications that need a different OS (e.g. Windows programs on a Mac).
- Running legacy software that does not work on a modern OS (e.g. an old Windows XP application).
- Security: safely opening suspicious files, malware analysis in an isolated VM with snapshots, and penetration-testing distributions (Kali Linux).
- Demos, training, proofs of concept and quickly trying out new systems.

*Table 2. Advantages and disadvantages of Type 2 hypervisors.*

| Advantages | Disadvantages |
|---|---|
| • Very easy to install and use (graphical interface)<br>• Works on almost any computer – wide hardware compatibility<br>• No dedicated machine needed; host and guests work at the same time<br>• Mostly free (VirtualBox, VMware Workstation Pro/Fusion Pro, QEMU)<br>• Convenient desktop integration (clipboard, shared folders, USB)<br>• Snapshots make experiments safe and repeatable | • Lower performance, especially for I/O; overhead from the host OS<br>• Security and stability depend on the host OS<br>• VMs compete with host applications for resources<br>• Poor scalability; no HA, clustering or live migration<br>• Possible conflicts with other hypervisors (e.g. Hyper-V on Windows)<br>• Not suitable for production server workloads |

## 4. Differences between Type 1 and Type 2 hypervisors

Table 3 summarises the main differences. The basic difference is the hypervisor's position in the software stack: **a Type 1 hypervisor controls the hardware itself, while a Type 2 hypervisor asks the host operating system for access to it**. Most of the other differences follow from this.

*Table 3. Comparison of Type 1 and Type 2 hypervisors.*

| Criterion | Type 1 (bare-metal) | Type 2 (hosted) |
|---|---|---|
| Position in the stack | Directly on the hardware; it is the OS for the VMs | Application or process running on a host OS |
| Hardware access | Direct (own drivers or a privileged partition such as dom0 or the root partition) | Indirect – through host OS drivers |
| Performance | Near-native, low overhead | Lower, especially I/O; extra layer and contention |
| Hardware compatibility | Limited – certified hardware (HCL) | Broad – anything the host OS supports |
| Security / isolation | High; small attack surface | Depends on the host OS; larger attack surface |
| Stability | Independent of a desktop OS; built for 24/7 | Host updates or crashes stop the VMs |
| Installation | Installed like an OS on a dedicated machine | Installed like an ordinary program |
| Management | Remote, centralised (web GUI, vCenter, SCVMM, APIs) | Local GUI on the same computer |
| Scalability | Tens to hundreds of VMs per host; clusters | A few VMs; single computer |
| Typical features | Live migration, HA, clustering, replication, passthrough, SR-IOV | Snapshots, shared folders, clipboard, USB, seamless mode |
| Cost | Often paid licence or subscription (free options exist) | Mostly free |
| Typical users | Data centres, cloud providers, companies | Students, developers, testers, home users |
| Examples | VMware ESXi, Microsoft Hyper-V, Xen, KVM* | Oracle VirtualBox, VMware Workstation and Fusion, Parallels Desktop, QEMU |

* **Borderline cases. KVM** (Kernel-based Virtual Machine) is a Linux kernel module. At first sight it looks like a Type 2 hypervisor because it is loaded into an ordinary Linux system. However, it turns the Linux kernel itself into a hypervisor, so many sources (e.g. Red Hat) classify it as Type 1. Others, including the article used in class, describe it as Type 2, because a full Linux system provides the virtualisation, not only the kernel [1][6]. **Microsoft Hyper-V** is the opposite case. It is installed as a Windows Server role or Windows feature, so it seems to be an application. However, enabling it places the hypervisor beneath Windows, and Windows then becomes the root partition running on top of it. Hyper-V is therefore a Type 1 hypervisor [1][5].

# 5. When to use which type

A simple rule is: **use Type 1 when virtual machines are the product** (production services that must be fast, secure and always available), and **use Type 2 when virtual machines are a**

**tool** (for learning, development, testing or occasionally running another OS on your own computer). Table 4 shows typical scenarios.

*Table 4. Recommended hypervisor type for typical scenarios.*

| Scenario | Recommended | Justification |
|---|---|---|
| Company production servers (AD, databases, mail, ERP) | Type 1 / server-designed platform | Performance, 24/7 reliability, HA, backup and central management |
| Cloud provider, hosting, VPS | Type 1 (KVM, Xen, Hyper-V) in a cloud platform | Very high VM density, multi-tenancy, automation through APIs |
| Virtual desktops (VDI) for employees | Type 1 / server-designed platform | Many desktops from central servers; GPU sharing |
| Consolidating old physical servers | Type 1 | Fewer machines, lower power and cooling costs, easier management |
| Small branch office or edge site (2–3 servers) | Type 1 (Hyper-V, Proxmox VE, small HCI) | Reliability with a small hardware footprint |
| Student running Linux on a Windows laptop for OS classes | Type 2 (VirtualBox, VMware Workstation) | Free, quick to install, host stays usable; snapshots allow safe experiments |
| Developer testing an application on many OSes | Type 2 | Quickly creating, cloning and resetting test VMs |
| Mac user who needs Windows applications | Type 2 (Parallels Desktop, VMware Fusion) | Good integration with the macOS desktop |
| Analysing suspicious files or malware | Type 2 with snapshots and an isolated network (dedicated Type 1 host for a professional lab) | Easy rollback; the threat is kept away from the host |
| Running old (legacy) software | Type 2 | Old OS in a VM without dedicating a whole machine |
| Emulating another CPU architecture (ARM, RISC-V) | QEMU (Type 2 emulator) | Only an emulator can run code for a different instruction set |
| Home lab for learning server administration | Server-designed (Proxmox VE, XCP-ng) on a spare PC | Learning clustering, HA and storage like in a real company, at no cost |

# 6. Server-designed virtualisers

## 6.1. Definition

**Server-designed virtualisers** (server-class or enterprise virtualisation platforms) are complete virtualisation solutions designed from the start to run **production workloads on servers and in data centres**, typically on **many physical hosts managed as one pool of resources**. In addition to the hypervisor they provide a whole ecosystem: central management, clustering, high availability, live migration, shared or software-defined storage and networking, backup, monitoring, user roles and permissions, and an API for automation.

This category is **not a third position in the stack** next to Type 1 and Type 2. It describes the **purpose and scope** of a product, not where the hypervisor sits. In practice almost every

server-designed platform is built on a **Type 1 hypervisor** (ESXi, Hyper-V, Xen or KVM). The difference is that a Type 1 hypervisor alone virtualises a single machine, while a server-designed platform manages a whole infrastructure. For example, ESXi is the hypervisor, and vSphere (ESXi + vCenter) is the server platform.

## 6.2. Characteristic features

- **Central management** of many hosts and VMs from one console (vCenter, System Center VMM / Windows Admin Center, Proxmox web GUI, Xen Orchestra, Nutanix Prism).
- **Clusters and high availability (HA)**: when a host fails, its VMs restart automatically on the remaining hosts.
- **Live migration** of running VMs and their storage between hosts, e.g. for hardware maintenance without downtime.
- **Automatic load balancing** (e.g. VMware DRS) and resource pools with quotas.
- **Software-defined storage and hyperconvergence (HCI)**: vSAN, Storage Spaces Direct, Ceph or ZFS – local disks of many servers combined into shared, redundant storage.
- **Software-defined networking (SDN)**: virtual switches, VLANs and VXLANs, microsegmentation, firewalls.
- **Backup, replication and disaster recovery** integration; templates and fast provisioning.
- **Security and compliance**: role-based access control (RBAC), audit logs, encrypted or shielded VMs, TPM emulation and Secure Boot.
- **Automation**: REST APIs, PowerShell/PowerCLI, Terraform, Ansible; often vendor support and certifications.

*Table 5. Advantages and disadvantages of server-designed virtualisers.*

| Advantages | Disadvantages |
|---|---|
| • Highest availability – HA and live migration minimise downtime<br>• Efficient use of the whole infrastructure (consolidation, load balancing)<br>• Central management and automation of hundreds of VMs<br>• Built-in or integrated storage, networking, backup and DR<br>• Vendor support, certifications and a large ecosystem<br>• Scales from a few servers to a whole data centre | • Highest cost (licences or subscriptions, often per CPU core) for commercial products<br>• Requires several servers, shared storage and fast networks for full functionality<br>• High complexity; requires qualified administrators<br>• Risk of vendor lock-in (e.g. the VMware licensing changes after 2023)<br>• Too much for a single computer or a home user (except open-source platforms used in home labs) |

### 6.3. When to use

Server-designed platforms are the right choice whenever a company runs **production services** that must be available all the time and are spread across **more than one physical server**. Examples are company data centres, private clouds, hosting, VDI and branch offices that need high availability. Smaller organisations and home labs increasingly use open-source platforms such as Proxmox VE or XCP-ng, which offer clustering and HA without licence fees.

## 7. Overview of virtualisers available on the market

The following sections describe the most popular products, divided into three groups, as the exercise requires. Group A contains the Type 1 hypervisors themselves, group B contains Type 2 (hosted) products, and group C contains complete server-designed platforms, most of which are built on the hypervisors from group A.

### 7.1. Group A – Type 1 hypervisors

**VMware ESXi (Broadcom)**

ESXi is a mature, widely used bare-metal hypervisor with its own small kernel (VMkernel). It installs on a server, is managed through a web-based Host Client or centrally through vCenter, and supports very large VMs, memory overcommitment, PCI passthrough and vGPU. **Capabilities:** very high performance and stability, and a huge ecosystem of backup and monitoring tools. **Limitations:** strict hardware compatibility list. After Broadcom acquired VMware in 2023, licensing changed to subscription bundles, which significantly raised costs for many customers. **Use:** enterprise data centres, as the basis of the vSphere platform (group C).

**Microsoft Hyper-V**

Hyper-V is Microsoft's Type 1 hypervisor, available as a role in Windows Server and as an optional feature of Windows 10/11 Pro, Enterprise and Education. It requires a 64-bit CPU with SLAT and hardware virtualisation enabled. Windows runs in the root partition, and VMs run in child partitions. **Capabilities:** Generation 1 and Generation 2 VMs (UEFI, Secure Boot, virtual TPM), checkpoints, Live Migration, Hyper-V Replica, nested virtualisation, Discrete Device Assignment and GPU partitioning. It is also the basis of Microsoft Azure, WSL 2, Windows Sandbox and virtualisation-based security. **Limitations:** management is Windows-oriented (Hyper-V Manager, PowerShell, Windows Admin Center, SCVMM), the free stand-alone Hyper-V Server was discontinued after the 2019 version, and the Windows

Home edition does not include it. **Use:** Windows-based companies, labs and developer workstations. It is the main subject of this course's laboratory.

**Xen Project (Linux Foundation)**

Xen is an open-source microkernel hypervisor created at the University of Cambridge (first release 2003). The hypervisor itself is very small. A privileged domain, **dom0** (usually Linux), provides drivers and management, and guests run as **domU** domains. Xen pioneered **paravirtualisation (PV)**, and also supports HVM and PVH guests. **Capabilities:** strong isolation and a small trusted code base, and it is also used in embedded and automotive systems. **Limitations:** configuration without a management platform is demanding. **Use:** AWS EC2 for many years, plus XenServer, XCP-ng and Qubes OS (a security-focused desktop OS).

**KVM – Kernel-based Virtual Machine**

KVM has been part of the Linux kernel since version 2.6.20 (2007). Loading the kvm module turns Linux into a hypervisor: every VM is an ordinary Linux process scheduled by the kernel. **QEMU** emulates the devices, **virtio** drivers give fast paravirtualised I/O, and **libvirt** / virt-manager / Cockpit provide management. **Capabilities:** open source (GPL), excellent performance, live migration, and it benefits from all Linux features (cgroups, SELinux/sVirt, filesystems). **Limitations:** on its own it has no central graphical management for many hosts and requires Linux knowledge. **Use:** the most widely used hypervisor in public clouds (Google Cloud, AWS Nitro, OpenStack). It is also the basis of Proxmox VE, Nutanix AHV and OpenShift Virtualization.

**IBM z/VM and PowerVM**

These are hypervisors for IBM mainframes (IBM Z) and IBM Power servers. z/VM descends from CP-67 and VM/370 (1960s–1970s), the first commercial virtual machine systems, which the Type 1 and Type 2 theory was originally based on. **Capabilities:** extreme reliability and thousands of Linux guests on one machine. **Limitations:** work only on IBM hardware and are very expensive. **Use:** banks, insurance companies and large enterprises.

## 7.2. Group B – Type 2 hypervisors

**Oracle VirtualBox**

VirtualBox is a free, open-source (GPLv2) hosted hypervisor for Windows, Linux and macOS hosts (including Apple Silicon since version 7.1). **Capabilities:** supports many guest systems,

snapshots, cloning, shared folders, NAT, bridged and internal networks, Guest Additions for desktop integration, and a command-line tool (VBoxManage). **Limitations:** lower 3D and I/O performance than commercial products, and the Extension Pack (USB 2/3 passthrough, disk encryption, etc.) is licensed only for personal and educational use. **Use:** the most popular choice for students, education, home users and quick tests.

**VMware Workstation Pro**

Workstation Pro is a professional Type 2 hypervisor for Windows and Linux. Since November 2024 it has been free for personal, educational and commercial use. **Capabilities:** very good performance and 3D support, snapshot trees, linked clones, a virtual network editor, virtual TPM and encrypted VMs, and the ability to connect to ESXi/vSphere and move VMs to them. **Limitations:** proprietary, and only runs on x86 hosts. **Use:** developers, IT trainers and administrators preparing VMs for vSphere.

**VMware Fusion Pro**

Fusion Pro is the macOS version of Workstation, also free since November 2024. On Apple Silicon Macs it runs ARM guests (Windows 11 on Arm, ARM Linux). **Capabilities:** Unity mode, which shows guest application windows directly on the macOS desktop, and good integration with the VMware ecosystem. **Limitations:** on Apple Silicon it cannot run x86 operating systems efficiently. **Use:** Mac users who need Windows or Linux.

**Parallels Desktop**

Parallels Desktop is a commercial hypervisor for macOS. It uses Apple's Hypervisor.framework, and Microsoft has officially approved it for running Windows 11 on Arm on Apple M-series computers. **Capabilities:** very good performance and graphics, Coherence mode (Windows applications as if they were native Mac applications), and one-click installation of Windows, Linux and macOS guests. **Limitations:** paid subscription and macOS only. **Use:** Mac users running Windows software (office, engineering or games).

**QEMU (and front-ends such as UTM)**

QEMU is an open-source machine emulator and virtualiser created by Fabrice Bellard in 2003. Without acceleration it works as a pure Type 2 **emulator** that uses dynamic binary translation (TCG), so it can run code for a **different CPU architecture** (ARM, RISC-V, PowerPC, MIPS) on an x86 PC. With acceleration (KVM on Linux, HVF on macOS, WHPX on Windows) it runs guests at near-native speed. **Limitations:** pure emulation is slow, and the command line is complex (front-ends such as virt-manager or UTM on macOS/iOS help).

**Use:** embedded-systems and OS-kernel development, testing other architectures, CI pipelines, and as the device-emulation component of KVM.

## 7.3. Group C – Server-designed virtualisers

**VMware vSphere / VMware Cloud Foundation (Broadcom)**

vSphere consists of ESXi hosts plus **vCenter Server**, the central management system. It is extended in VMware Cloud Foundation with vSAN (hyperconverged storage), NSX (network virtualisation) and automation tools. **Capabilities:** vMotion and Storage vMotion, HA, DRS load balancing, Fault Tolerance (a shadow copy of a VM running in lockstep), distributed switches, and a huge ecosystem. **Limitations:** high cost after the subscription-only licensing change in 2024, and vendor lock-in. **Use:** large enterprises, banks, public administration and service providers. It has long been the market reference point.

**Microsoft Hyper-V on Windows Server and Azure Local**

Hyper-V hosts combined with **Failover Clustering**, **Storage Spaces Direct** and management through System Center Virtual Machine Manager or Windows Admin Center form a complete server platform. **Azure Local** (formerly Azure Stack HCI) is Microsoft's hyperconverged system managed from the Azure cloud. **Capabilities:** clustered HA, Live Migration, Hyper-V Replica, SDN, shielded VMs, and close integration with Active Directory and Azure. **Use:** organisations based on Microsoft technologies, and hybrid cloud.

**Proxmox Virtual Environment (Proxmox VE)**

Proxmox VE is an open-source (AGPLv3) server platform based on Debian Linux, from Proxmox Server Solutions GmbH (Vienna). It combines **KVM** virtual machines and **LXC** containers and is managed through a web interface. **Capabilities:** multi-node clusters, HA, live migration, built-in **Ceph** and **ZFS** storage, a firewall, SDN, and integration with Proxmox Backup Server. **Limitations:** the stable enterprise repository and official support require a paid subscription, and it has a smaller ecosystem than VMware. **Use:** small and medium companies, schools and home labs. It is a popular alternative to VMware after 2024.

**XenServer and XCP-ng**

XenServer (formerly Citrix Hypervisor, Cloud Software Group) and its community fork **XCP-ng** (Vates) are server platforms based on the **Xen** hypervisor. XCP-ng is managed with **Xen Orchestra**. **Capabilities:** resource pools, HA, live migration including storage, backup, and strong Citrix VDI integration (XenServer). **Limitations:** a smaller community than

KVM-based solutions. **Use:** VDI (Citrix Virtual Apps and Desktops), and companies looking for an open-source replacement for vSphere.

**Nutanix AHV**

AHV is a KVM-based hypervisor included in the **hyperconverged** Nutanix Cloud Infrastructure, managed through **Prism**. Compute and distributed storage run on the same nodes. **Capabilities:** simple one-click upgrades, HA, built-in replication and DR, and no separate hypervisor licence. **Limitations:** tied to the Nutanix software stack and certified hardware. **Use:** companies that want a simple, scalable private cloud. It is often chosen when migrating away from VMware.

**OpenStack and OpenShift Virtualization (KubeVirt)**

**OpenStack** is an open-source cloud platform (IaaS). Its Nova component manages hypervisors, most often KVM, across hundreds or thousands of servers, with self-service, multi-tenancy and APIs. **Red Hat OpenShift Virtualization**, based on the KubeVirt project, runs VMs next to containers inside Kubernetes. It is Red Hat's successor to the discontinued Red Hat Virtualization (oVirt). **Limitations:** both are very complex and require large teams to operate. **Use:** telecoms, research centres, public and private clouds, and companies modernising applications towards containers.

*Table 6. Summary of the virtualisers described.*

| Product | Group / type | Licence | Host / platform | Main use |
|---|---|---|---|---|
| VMware ESXi | A – Type 1 | Proprietary | Bare-metal x86 server | Enterprise servers |
| Microsoft Hyper-V | A – Type 1 | Part of Windows / Windows Server | Bare metal (Windows in root partition) | Windows servers, labs, developers |
| Xen Project | A – Type 1 | Open source (GPLv2) | Bare metal + dom0 | Clouds, security, embedded |
| KVM | A – Type 1* | Open source (GPL) | Linux kernel | Public clouds, basis of many platforms |
| IBM z/VM, PowerVM | A – Type 1 | Proprietary | IBM Z / Power | Mainframes, large enterprises |
| Oracle VirtualBox | B – Type 2 | Open source + PUEL extension pack | Windows, Linux, macOS | Education, home users, tests |
| VMware Workstation Pro | B – Type 2 | Proprietary, free | Windows, Linux | Developers, trainers |
| VMware Fusion Pro | B – Type 2 | Proprietary, free | macOS | Mac users |
| Parallels Desktop | B – Type 2 | Commercial subscription | macOS | Windows apps on Mac |
| QEMU / UTM | B – Type 2 (emulator) | Open source | Linux, Windows, macOS | Cross-architecture emulation |

| Product | Group / type | Licence | Host / platform | Main use |
|---|---|---|---|---|
| VMware vSphere / VCF | C – Server (ESXi) | Subscription | Server clusters | Enterprise data centres |
| Hyper-V + Failover Clustering / Azure Local | C – Server (Hyper-V) | Windows Server / Azure subscription | Server clusters, HCI | Microsoft environments, hybrid cloud |
| Proxmox VE | C – Server (KVM + LXC) | Open source, optional subscription | Server clusters | SMEs, schools, home labs |
| XenServer / XCP-ng | C – Server (Xen) | Commercial / open source | Server pools | VDI, vSphere alternative |
| Nutanix AHV | C – Server (KVM, HCI) | Commercial | Nutanix HCI nodes | Private cloud |
| OpenStack, OpenShift Virtualization | C – Server / cloud (KVM) | Open source / commercial | Large clusters | Private and public clouds |

# 8. Conclusions

Type 1 and Type 2 hypervisors are not competitors but tools for different jobs. A **Type 1** hypervisor runs directly on the hardware and offers the best performance, security and scalability, so it is the standard in data centres and clouds. Its costs are limited hardware support and more complex administration. A **Type 2** hypervisor is an application on an ordinary operating system. It is easy to install, free in most cases and works on almost any computer, which makes it ideal for learning, development and testing. It pays for this with lower performance and a dependence on the host OS.

**Server-designed** virtualisers are complete platforms built around a Type 1 hypervisor. They move virtualisation from a single machine to the whole infrastructure, adding central management, clustering, high availability, live migration, and software-defined storage and networking. The boundaries between the categories are blurring: KVM and Hyper-V look like applications but work as Type 1 hypervisors, and Type 2 products such as VMware Workstation can connect to server platforms. The choice should therefore depend on what the virtual machines are for: **a tool on your own computer (Type 2)**, **a production service on a dedicated server (Type 1)** or **an entire infrastructure (server-designed platform)**.

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